

Multiplying and Dividing Fractions: Find the Flip Mistake

Cancelling correctly before multiplying, flipping only the divisor, and finding and fixing a misused reciprocal

Two rules this page is about.

R1. Multiplying: multiply straight across. You may cancel a *top* number with a *bottom* number first — never a top with a top, and never a bottom with a bottom.

R2. Dividing: keep the first fraction, change $\div$ to $\times$, and flip *only the second* fraction (the divisor). Cancel **after** the flip, never before.

For every “find the mistake” problem, name the mistake in words *and* show correct work. Leave answers as fractions (improper fractions are fine).

1. Multiply. Cancel a top number with a bottom number first, then multiply straight across.

$$\frac{2}{3} \times \frac{3}{8}$$

Answer: _____

2. Divide. Write down which fraction you flipped before you compute, and why that is the one that flips.

$$\frac{3}{4} \div \frac{2}{5}$$

Answer: _____

3. Find and fix. Kai divided like this:

$$\frac{2}{3} \div \frac{4}{5} = \frac{3}{2} \times \frac{4}{5} = \frac{6}{5}$$

(a) Name Kai's mistake in words. (b) Show the correct work and give the correct answer.

Answer: _____

4. Find and fix. Maya multiplied like this, saying she "cancelled the 4 and the 8":

$$\frac{4}{5} \times \frac{8}{9} = \frac{1}{5} \times \frac{2}{9} = \frac{2}{45}$$

(a) Explain why that cancelling is not allowed. (b) Give the correct product.

Answer: _____

5. Find the missing factor. To undo multiplying by $\frac{3}{4}$ you divide by $\frac{3}{4}$, which means multiplying by its reciprocal $\frac{4}{3}$.

$$\frac{3}{4} \times \underline{\hspace{2cm}} = \frac{1}{2}$$

6. Find and fix. Jordan cancelled before flipping:

$$\frac{6}{7} \div \frac{3}{8} = \frac{3}{7} \div \frac{3}{4} = \frac{3}{7} \times \frac{4}{3} = \frac{4}{7}$$

(a) Explain why no cancelling is allowed until the division has been rewritten as multiplication. (b) Give the correct answer.

Answer: _____

7. Who is right? Ali and Sam each computed $\frac{5}{6} \div \frac{5}{8}$.

Ali: $\frac{6}{5} \times \frac{5}{8} = \frac{3}{4}$ Sam: $\frac{5}{6} \times \frac{8}{5} = \frac{4}{3}$

Which student took the reciprocal of the right fraction? Name the other student's mistake, and write the correct answer.

Answer: _____

8. Find and fix (word problem). One batch of granola uses $\frac{3}{4}$ cup of oats. You have $\frac{9}{2}$ cups of oats. How many whole batches can you make?

Nia wrote $\frac{2}{9} \times \frac{3}{4} = \frac{1}{6}$ batch. Explain why her answer cannot be right (you have more oats than one batch needs), then give the correct number of batches.

Answer: _____ batches